

# Dhawal Sirikonda

Ph.D. Candidate • Rendering & Imaging Science Lab (RISc), Dartmouth College

✉ dhawal.sirikonda.gr@dartmouth.edu | 🌐 dhawal.xyz | 🔄 dhawals1939 | 🔍 google-scholar | 📍 Hanover, NH, USA

## 🔬 research summary

I build next-generation, low-cost LiDAR and computational imaging systems for high-speed perception and communication. I design sensing architectures that reach MHz-rate depth measurements in time-of-flight sensors, and use acousto-optic lensing for structured-light scanning ( $\sim 1000\times$  faster than prior art) and underwater optical backscatter communication ( $\sim 600\times$  faster). I develop physics-driven methods across event-camera, RGB-D, and emerging hybrid sensing modalities. I am advised by [Prof. Adithya Pediredla](#). Before my Ph.D., I completed an M.S. at IIIT-Hyderabad with [Prof. P.J. Narayanan](#) on computer graphics and 3D vision.

## 🎓 education

|                     |                                                                                                 |                         |
|---------------------|-------------------------------------------------------------------------------------------------|-------------------------|
| Sep 2023 – Present  | Ph.D., Computer Science — Dartmouth College                                                     | Hanover, NH, USA        |
|                     | Rendering & Imaging Science Lab (RISc) • Advisor: Prof. Adithya Pediredla                       |                         |
| Jan 2020 – May 2023 | M.S., Computer Science — IIIT-Hyderabad                                                         | Hyderabad, TS, India    |
|                     | Center for Visual Information Technology (CVIT) • Advisor: Prof. P.J. Narayanan                 |                         |
|                     | Thesis: Real-Time Rendering of Arbitrary Surface Geometries using Precomputed Radiance Transfer |                         |
| Aug 2019 – Dec 2019 | M.Tech., Computer Science (discontinued) — IIIT-Hyderabad                                       | Hyderabad, TS, India    |
|                     | Discontinued after one semester to transfer into the M.S. by Research program                   |                         |
| 2014 – 2018         | B.Tech., Computer Science — JNTUK – University College of Engineering                           | Vizianagaram, AP, India |
|                     | Academically 2nd in the 2014–2018 batch (Roll of Honours)                                       |                         |

## 📖 publications

- [1] [SDR-ToF: Million Optical Depth Samples per Second Using Software-Defined Radio](#) *SIGGRAPH Asia (ACM ToG)*, 2026.  
Ramchander Bhaskara\*, **Dhawal Sirikonda\***, Omkar Vengurlekar\*, Juhyeon Kim, Joseph Lazarro, Suren Jayasuriya, Adithya Pediredla (\*  $\equiv$  joint first authors)
- [2] [Underwater Optical Backscatter Communications using Acousto-Optic Beam Steering](#) *SIGGRAPH Asia (ACM ToG)*, 2025.  
Atul Rohit Agarwal\*, **Dhawal Sirikonda\***, Atharv Agashe, Ziang Ren, Dinithi Silva-Sassaman, Charles Carver, Alberto Quattrini Li, Xia Zhou, Adithya Pediredla (\*  $\equiv$  joint first author)
- [3] [Structured light with a million light planes per second](#) *ICCP & TPAMI*, 2025.  
**Dhawal Sirikonda**, Praneeth Chakravarthula, Ioannis Gkioulekas, Adithya Pediredla
- [4] [GSN: Generalisable Segmentation in Neural Radiance Field](#) *AAAI*, 2024.  
Vinayak Gupta, Rahul Goel, **Dhawal Sirikonda**, P.J. Narayanan
- [5] [FusedRF: Fusing Multiple Radiance Fields](#) *CVPR Workshop (XRNeRF)*, 2023.  
Rahul Goel, **Dhawal Sirikonda**, Rajvi Shah, P.J. Narayanan
- [6] [Interactive Segmentation of Radiance Fields](#) *CVPR*, 2023.  
Rahul Goel\*, **Dhawal Sirikonda\***, Saurabh Saini, P.J. Narayanan (\*  $\equiv$  joint first author)
- [7] [Real-Time Rendering of Arbitrary Surface Geometries using Learnt Transfer](#) *ICVGIP*, 2022.  
**Dhawal Sirikonda**, Aakash KT, P.J. Narayanan
- [8] [StyleTRF: Stylizing Tensorial Radiance Fields](#) *ICVGIP*, 2022.  
Rahul Goel\*, **Dhawal Sirikonda\***, Saurabh Saini, P.J. Narayanan (\*  $\equiv$  joint first author)
- [9] [Neural View Synthesis with Appearance Editing from Unstructured Images](#) *ICVGIP*, 2021.  
Pulkit Gera, Aakash KT, **Dhawal Sirikonda**, P.J. Narayanan

## preprints & posters

- › [Learnt Transfer for Surface Geometries](#) HPG 2022 (Poster)
- › [Transfer Textures for Fast Precomputed Radiance Transfer](#) EuroGraphics 2022 (Poster)
- › [PRTT: Precomputed Radiance Transfer Textures](#) arXiv 2022
- › [Appearance Editing with Free-viewpoint Neural Rendering](#) arXiv 2021

## 🌱 patents

- › [MHz-Rate Optical Depth Imaging Using Software-Defined Radios](#)  
U.S. Patent App. No. 64/113,861, filed Jul. 17, 2026 • Pediredla et al.
- › [Underwater Optical Backscatter Communication Using Acousto-Optic Beam Steering](#)  
U.S. Patent App. No. 63/913,085, filed Nov. 6, 2025 • Pediredla et al.
- › [Event-Enhanced Histopathology Apparatus](#)  
U.S. Patent App. No. 63/837,253, filed Jul. 2, 2025 • Pediredla et al.

## 🎤 talks & presentations

- › [Using Sound to Steer Light and Light to Measure Sound](#) VASC Seminar, Carnegie Mellon University, Pittsburgh, PA, USA • 2026
- › [\[live demo\] Controlling Light With Sound](#) ICCP 2026, Princeton, NJ, USA • 2026
- › [From Rendering to Imaging: Controlling Light with Sound](#) NESG 2026, New England Symposium on Graphics, MIT, Cambridge, MA, USA • 2026
- › [Underwater Optical Backscatter Communications using Acousto-Optic Beam Steering](#) SIGGRAPH Asia 2025, Hong Kong SAR • 2025
- › [Structured light with a million light planes per second](#) NECV 2025, University of Massachusetts Amherst, MA, USA • 2025
- › [Structured light with a million light planes per second](#) ICCP 2025, Toronto, ON, Canada • 2025
- › [Interactive Segmentation of Radiance Fields](#) NECV 2023, Dartmouth College, Hanover, NH, USA • 2023

## 📁 experience

|                       |                                                                                                                                                                                                                                                |                      |
|-----------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------|
| Sep 2023 – Present    | <b>Ph.D. Researcher</b><br><i>Rendering &amp; Imaging Science Lab, Dartmouth College</i><br>Research on acousto-optic imaging for ultra-fast scanning, optical communication, and acoustic lensing; low-cost LiDAR from commodity RF hardware. | Hanover, NH, USA     |
| June 2026 – Sept 2026 | <b>Machine Learning Engineer</b><br><i>Bosch Research</i>                                                                                                                                                                                      | Pittsburgh, PA, USA  |
| Jan 2020 – May 2023   | <b>Research Assistant</b><br><i>Center for Visual Information Technology, IIIT-Hyderabad</i><br>Differentiable rendering and multi-view surface reconstruction (Mitsuba 2); supervised undergraduate and dual-degree students.                 | Hyderabad, TS, India |

## 👩🏫 teaching

### Dartmouth College | Teaching Assistant

|                           |                    |
|---------------------------|--------------------|
| Computational Photography | Spring 2024 & 2025 |
| Computer Vision           | Winter 2024        |

### IIIT-Hyderabad | Teaching Assistant

|                           |             |
|---------------------------|-------------|
| Advanced Graphics (AR/VR) | Fall 2022   |
| Computer Graphics         | Spring 2021 |

### TalentSprint | Mentor — AI/ML program (with IIIT-Hyderabad)

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|---------------------------------------------|-------------|
| Capstone projects for working professionals | 2022 – 2023 |
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## mentoring

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- › Ava Carlson Undergraduate, Dartmouth College 2025 – Present
- › Saurish Gali High-school student, Oakton High School 2025
- › Atul R. Agarwal Graduate student, Dartmouth College 2024 – 2025
- › Atharv Agashe Undergraduate, Dartmouth College 2024 – 2025
- › Rahul Goel Undergraduate, IIIT-Hyderabad 2022 – 2023

## honours & service

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- › **Reviewer:** ICVGIP 2023; AAAI 2026; CVPR 2026; SIGGRAPH 2026; NeurIPS 2026 2023 – Present
- › **Roll of Honours** — academically 2nd in the 2014–2018 batch, JNTUK–UCEV 2018
- › Certified Programmer in Building Systems and Applications, MissionRnD 2016 – 2017

## outreach

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2024 – 2026      Dartmouth Science Day  
Demos to spark grade 2–10 students’ curiosity about imaging science; Hanover, NH.

## technical skills

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**Programming** Python, C/C++, MATLAB

**Libraries & Frameworks** PyTorch, CUDA, OptiX, OpenGL, Mitsuba 2

**Focus areas** Computational imaging, physically based rendering, advanced graphics (AR/VR), computer vision, statistical methods in AI